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$$\begin{aligned} & \int \frac{dx}{\sqrt{x(4-x)}} \\ &= \int \frac{dx}{\sqrt{-x^2 + 4x}} \\ &\Rightarrow -x^2 + 4x = -(x^2 - 4x + 2^2) + 4 = 4 - (x-2)^2 \\ &\Rightarrow \int \frac{dx}{\sqrt{4 - (x-2)^2}} \\ &\text{Stel } t = x - 2 \Rightarrow dt = dx \\ &\Rightarrow \int \frac{dt}{\sqrt{4 - t^2}} = \int \frac{dt}{\sqrt{2^2 - t^2}} \\ &= \text{Bgsin} \frac{t}{2} + c = \text{Bgsin} \frac{x-2}{2} + c \end{aligned}$$

References